



Module Presenter's Manual

for

Integrating Applications with
Spring Framework

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Ver. 1.0

Amendment Record

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1.0	June 2024	New	-

Table of Contents

Sr. No.	Details	Page No.
1.	Introduction	1
2.	Information on Session Allocation	2
3.	Module Deliverables on OnlineVarsity	2
4.	Week-wise Session Schedule	3
5.	Session Coverage	4
6.	Library References	7

1. Introduction

At the end of this course, students will be able to:

- Learn the basics of Spring Framework 6.0
- Understand Spring Modules and Projects
- Understand Dependency Injection
- Learn XML Configuration for Spring Applications
- Understand Spring MVC
- Learn the advanced features of Spring
- Integrate Spring MVC with Bootstrap
- Add Spring Security Dependency
- Learn Microservices and Cloud Native Applications
- Learn Spring Boot and Spring Cloud
- Understand Microservices with Spring Boot
- Use Spring Data
- Learn Spring Security and Validations

2. Information on Session Allocation

Module	Online Hours
Integrating Applications with Spring Framework	16

Throughout this Presenter's Manual, the module **Integrating Applications with Spring Framework** will be referred to as **Spring**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables:

Resources available on Onlinevarsity:

Feature - Description/Functionality
Ebook -Student reading material in the form of PDF.
Download eBook - Student has the option to download the subject related e-book and read offline.
Practice 4 Me - Student can test and evaluate their understanding of module related topics.
Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
References - Student can access additional subject related material for reading.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 SPRING – TL1	Session 2 SPRING – TL2	Session 3 SPRING – TL3	Session 4 SPRING – TL4
2	Session 5 SPRING– TL5	Session 6 SPRING – TL6	Session 7 SPRING – TL7	Session 8 SPRING – TL8

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	SPRING – TL1	All the topics as listed below from Session 1 of <i>Spring and Spring Boot with Jakarta EE Web Applications</i> book should be covered in this session. <u>Session 1: Introducing the Spring Framework</u> ➤ Explain the features and benefits of the Spring Framework ➤ Describe the Spring Framework Architecture ➤ Identify the main use cases of the Spring Framework ➤ List the steps to install Spring Framework 6.0	<u>Spring and Spring Boot with Jakarta EE Web Applications</u> SG - Session 1 XP - Session 1 TG - Session 1
2	SPRING – TL2	The following topics as listed below from Session 2 of <i>Spring and Spring Boot with Jakarta EE Web Applications</i> book should be covered in this session. <u>Session 2: Creating Web Applications Using Spring MVC</u> ➤ Describe the Spring MVC architecture and key components ➤ Explain how to create controllers, views, and models for Web applications ➤ Describe how to handle HTTP requests and responses with Spring MVC ➤ List the annotations in Spring Validation and how to add validations in applications ➤ List the steps to integrate Spring MVC with Jakarta EE Web environment	<u>Spring and Spring Boot with Jakarta EE Web Applications</u> SG - Session 2 XP - Session 2 TG - Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
3	SPRING – TL3	All the topics as listed below from Session 3 of <i>Spring and Spring Boot with Jakarta EE Web Applications</i> book should be covered in this session. <u>Session 3: Working with Spring Boot and Spring Data</u> ➤ Identify the purpose of Spring Boot ➤ Describe the components of the Spring Boot architecture ➤ Explain the JPA architecture ➤ Outline the steps to create a Spring Boot application with MySQL database	<u>Spring and Spring Boot with Jakarta EE Web Applications</u> SG - Session 3 XP - Session 3 TG - Session 3
4	SPRING – TL4	The Try It Yourself questions of Session 1 to Session 3 of <i>Spring and Spring Boot with Jakarta EE Web Applications</i> book should be covered in this session.	<u>Spring and Spring Boot with Jakarta EE Web Applications</u> Session 1 to 3
5	SPRING – TL5	All the topics as listed below from Session 4 of <i>Spring and Spring Boot with Jakarta EE Web Applications</i> book should be covered in this session. <u>Session 4: Working with WebFlux</u> ➤ Describe reactive programming architecture ➤ Explain Spring WebFlux ➤ Distinguish between Spring MVC and Spring WebFlux ➤ Outline the steps to create an asynchronous program using WebFlux	<u>Spring and Spring Boot with Jakarta EE Web Applications</u> SG - Session 4 XP - Session 4 TG - Session 4

Session No.	Session Title	Session Details	Deliverables' Mapping
6	SPRING – TL6	The following topics as listed below from Session 5 of <i>Spring and Spring Boot with Jakarta EE Web Applications</i> book should be covered in this session. <u>Session 5: Creating Cloud-Native Applications</u> ➤ Explain the process of authenticating and authorizing users using Spring Security ➤ Describe how to secure Web resources and API endpoints ➤ Explain how to create cloud-native applications using Spring Cloud ➤ Deploy a Web application on the cloud	<u>Spring and Spring Boot with Jakarta EE Web Applications</u> SG - Session 5 XP - Session 5 TG - Session 5
7	SPRING – TL7	The following topics as listed below from Session 6 of <i>Spring and Spring Boot with Jakarta EE Web Applications</i> book should be covered in this session. <u>Session 6: Creating Microservices with Spring Boot</u> ➤ Describe the microservices architecture ➤ Explain the process of creating microservices using Spring Boot ➤ List the steps to deploy microservices on cloud platforms	<u>Spring and Spring Boot with Jakarta EE Web Applications</u> SG - Session 6 XP - Session 6 TG - Session 6
8	SPRING – TL8	The Try It Yourself questions of Session 4 to Session 6 of <i>Spring and Spring Boot with Jakarta EE Web Applications</i> book should be covered in this session.	<u>Spring and Spring Boot with Jakarta EE Web Applications</u> Session 4-6

6. Library References

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| ➤ Spring Boot 3 and Spring Framework 6 by Christian Ullenboom |
| ➤ Beginning Spring 6 From Beginner to Pro by Joseph B. Ottinger, Andrew Lombardi |

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